Subject-Level Calibration Heterogeneity is Stable Across LLM Families and Scales: A Statistical Audit of MMLU

Jennifer Zhang, John Wright
Department of Statistics, University of California, Berkeley

Motivation

- A model is **well-calibrated** if its stated confidence matches its accuracy (e.g., 80% confident ⇒ correct 80% of the time).
- Calibration is usually reported as one aggregate scalar (ECE), hiding *where* miscalibration concentrates:

$$\mathrm{ECE} = \sum_b \frac{|B_b|}{n} |\mathrm{acc}(B_b) - \mathrm{conf}(B_b)|$$

- **Questions:** where does miscalibration concentrate, and how much; is that pattern consistent across models; what drives it (question features, model scale)?

Data & Methods

- **Data:** MMLU — 14,042 four-option questions, 57 subjects, 4 domains; subject sizes 100–1,534 (median 173).
- **Confidence:** renormalized softmax over the four answer tokens:

$$c_i = \max_{a \in \{A,B,C,D\}} \frac{e^{\ell_a}}{\sum_{a'} e^{\ell_{a'}}}$$

(bounded below at 0.25 by construction — four-way softmax.)

- **Models:** GPT-4o-mini, Llama-3-8B-Instruct, Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. Outcome: signed *calibration gap* $\mathrm{gap}_i = c_i - y_i$, $y_i \in \{0,1\}$ correctness (positive under overconfidence).
- **Model:** three-level mixed model (questions → subjects → domains), fit via REML:

$$\mathrm{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}$$

u_k: domain random effect; v_{jk}: subject-within-domain random effect; ε_{ijk}: question-level residual.

- **Per-subject test:** one-sample t-test on question-level gaps (H_0: mean gap $= 0$), BH-FDR corrected across 57 subjects at $\alpha = 0.05$ (detailed next column).
- **Cross-model stability:** Spearman correlation of subject-level ECE across all 15 pairs among the 6 larger models, with bootstrap 95% CIs (detailed in Cross-Model Stability panel).

Structure Exists, But Is Modest

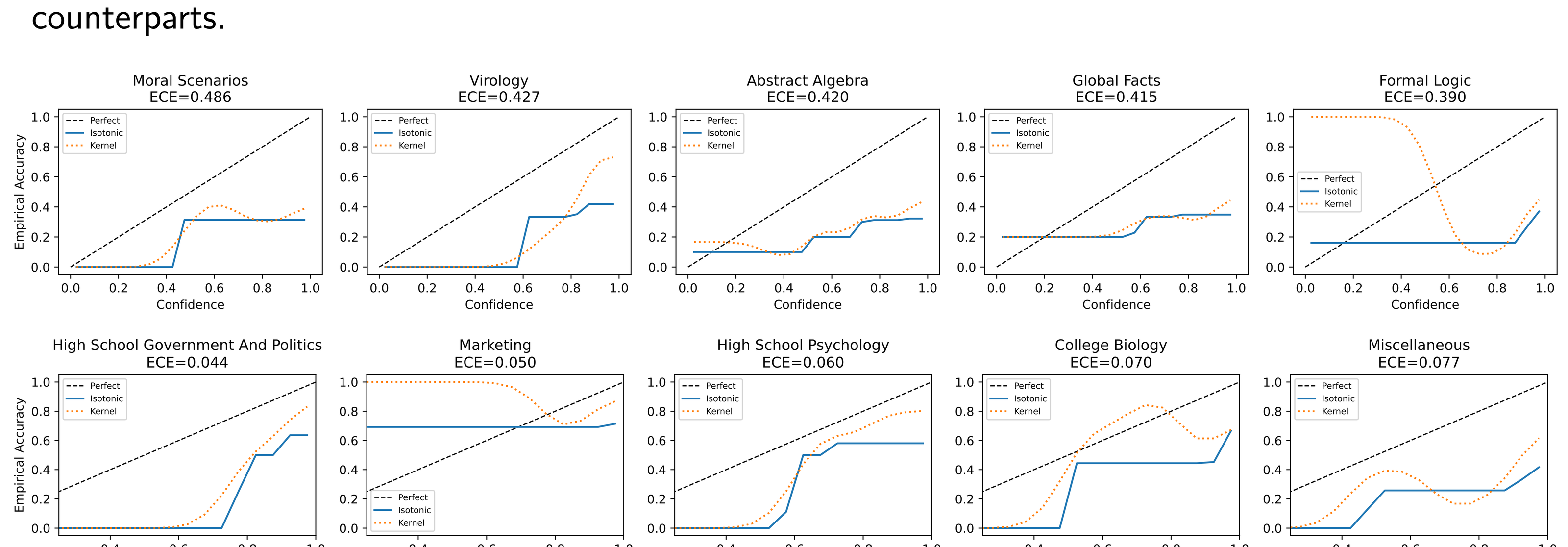
- Most calibration-gap variance is at the question level (92.0–98.5%); domain variance collapses to zero.
- Between-subject ICC spans nearly 5× across models — small in absolute terms, but (next block) large enough to organize a 3.8–13.0× range in subject-level miscalibration.

Model	$\mathrm{ICC}_{\mathrm{subject}}$
GPT-4o-mini	0.079
Llama-3-8B	0.020
Qwen-2.5-0.5B	0.018
Qwen-2.5-1.5B	0.017
Qwen-2.5-3B	0.051
Qwen-2.5-7B	0.049
Qwen-2.5-14B	0.059

$\mathrm{ICC}_{\mathrm{subject}}$: share of total calibration-gap variance attributable to subject (domain variance is estimated at zero for all models).

Where Calibration Fails

- **Every model is overconfident on average:** intercept (μ) positive for all seven models, 0.18–0.31 (Guo et al. 2017; Kadavath et al. 2022).
- **Worst, every model >1B:** `virology`, `professional_law` (top-10); college-STEM cluster — `abstract_algebra`, `college_mathematics`, `machine_learning` (top-5, 4 of 6 models).
- **Best, every model:** `marketing`, `high_school_psychology` (lowest ECE).
- **Pattern:** specialization level, not domain — college subjects miscalibrate more than their high-school counterparts.



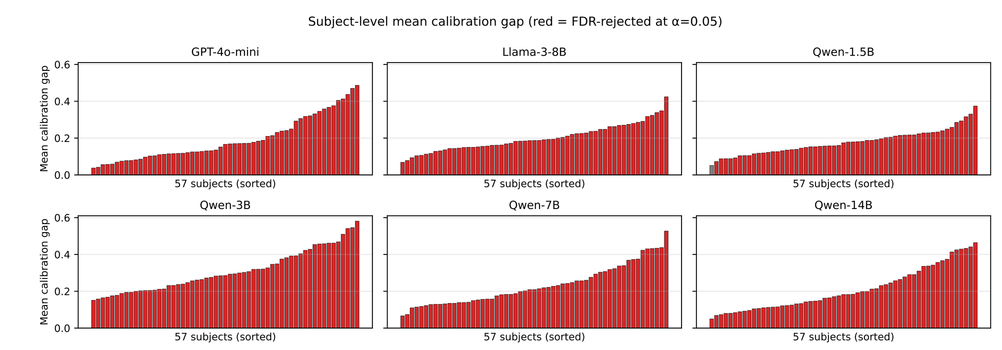
GPT-4o-mini's own five most miscalibrated (top) and five best-calibrated (bottom) subjects by ECE — the cross-model-robust worst/best list above differs slightly since it requires agreement across all seven models (see Cross-Model Stability). Solid: isotonic fit; dotted: kernel smoother; dashed: perfect calibration.

Magnitude, Not the Binary Count

- For each of the 57 subjects s, test whether its mean gap is zero:

$$H_0: \ \mathbb{E}[\mathrm{gap}_i \mid \text{subject } s] = 0 \quad \text{vs.} \quad H_1: \ \mathbb{E}[\mathrm{gap}_i \mid \text{subject } s] \neq 0,$$

- One-sample t-tests, FDR-controlled at $\alpha = 0.05$ (Benjamini–Hochberg, 1995). Rejections are near-universal (five of seven models reject all 57) — expected given the sample size, *not informative* on its own.

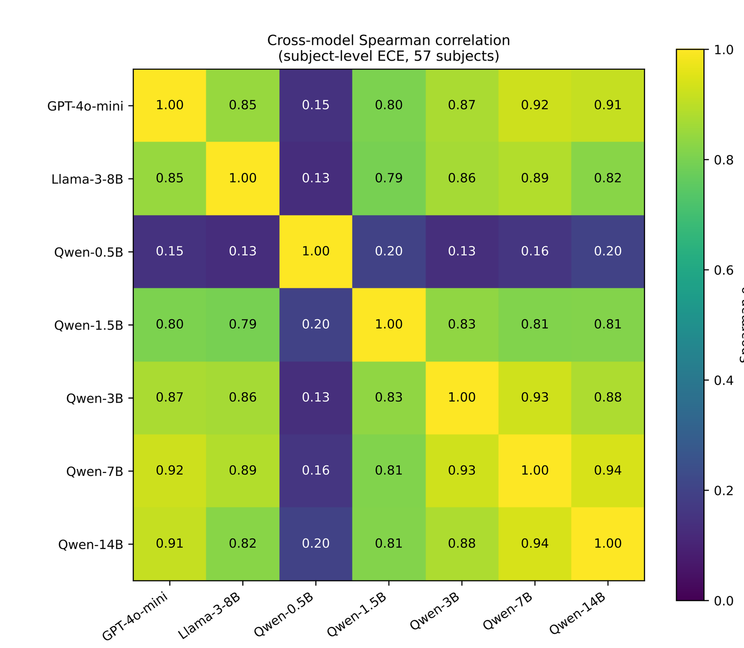


Mean calibration gap per subject, sorted ascending, per model. Red bars: FDR-rejected at $\alpha = 0.05$.

- The real finding is the *magnitude*: best- to worst-calibrated gap ranges span 3.8–13.0× across the six larger models, and the *same* subjects sit at each extreme, every time (robust to bin count and choice of scoring rule).

Stable Across Models and Families

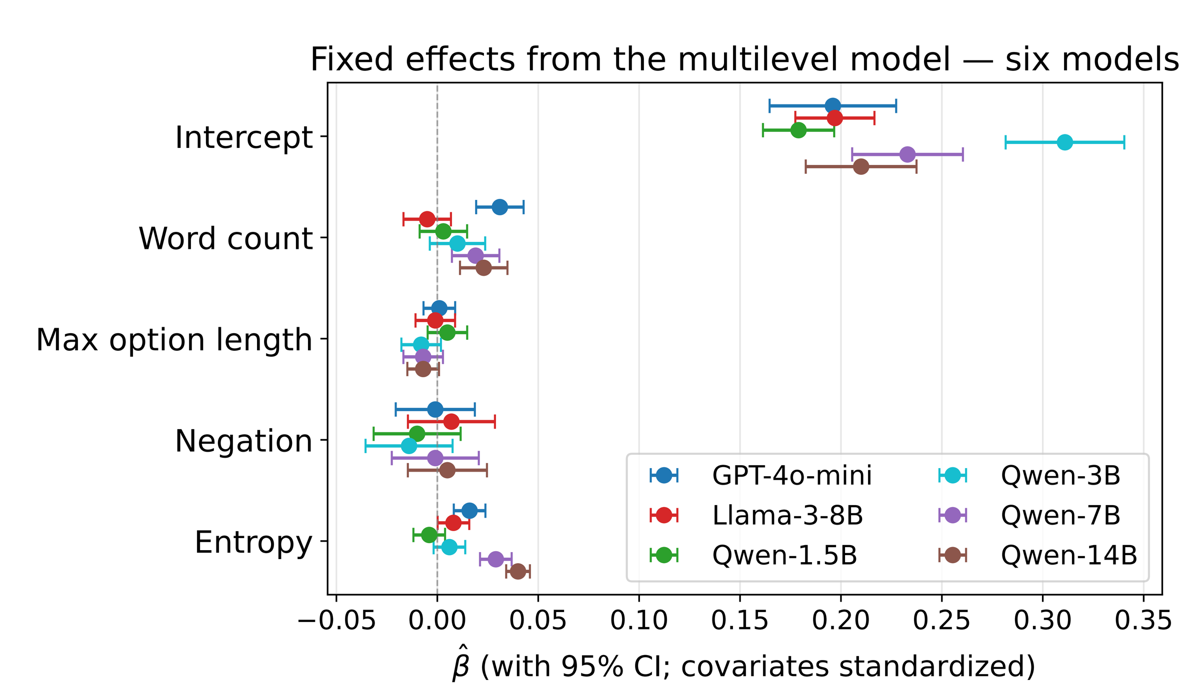
- All 15 pairwise Spearman correlations among the six larger models fall in [0.80, 0.94].
- Cross-family GPT-4o-mini vs. Qwen-7B/14B (0.92/0.91) is as strong as the best within-family pair (Qwen-7B vs. 14B, 0.94) — the same subjects are hard *regardless of architecture or scale*.
- Qwen-0.5B is a conspicuous exception ($\rho \in [0.13, 0.20]$, CIs include zero): it predicts D 59% of the time vs. A 5.6%, so its confidence tracks answer position, not item difficulty — its subject ranking isn't a meaningful calibration signal below ~1B parameters.



Spearman rank correlation of subject-level ECE across the seven models (57 common subjects).

What Predicts the Gap

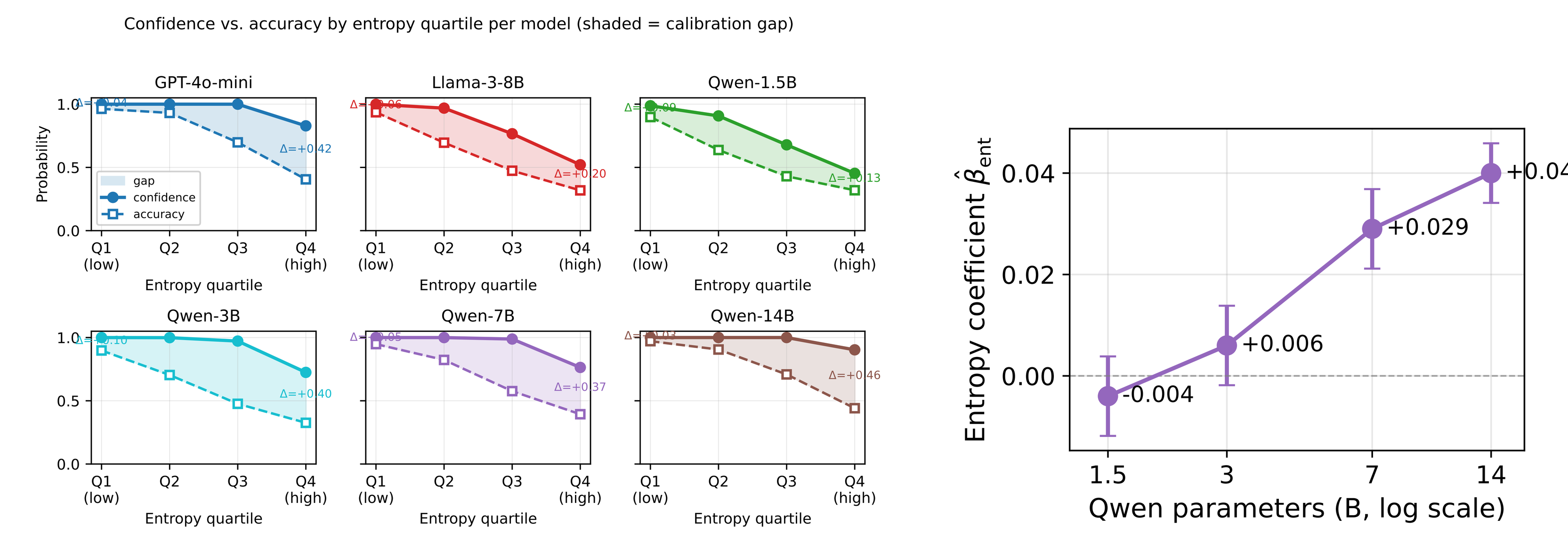
- Word count positively predicts overconfidence for GPT-4o-mini and the larger Qwens.
- Negation and option length: mostly inert.
- Most informative pattern: the **entropy** coefficient (next panel).



Fixed-effect estimates (±95% CI) on the standardized calibration-gap scale, six models (Qwen-0.5B excluded — positional bias, see Cross-Model Stability). The entropy coefficient shows a clear sign progression across the Qwen scaling sweep.

The Gap Grows With Scale

- **Naive prediction:** higher entropy (harder item) should mechanically lower confidence, shrinking the gap.
- **What happens instead:** confidence *does* drop as entropy rises, as predicted — but accuracy drops faster still, so the gap widens on the hardest items.
- **Gets worse with scale:** the entropy coefficient flips sign across the Qwen sweep (1.5B → 14B, right).



Left: confidence (solid) and accuracy (dashed) by entropy quartile, shaded = calibration gap. Right: entropy coefficient $\hat\beta_{\mathrm{ent}}$ across the Qwen scaling sweep (±95% CI).

Robustness Checks

- **ECE ranking:** $\rho \geq 0.97$ across bin counts 10–30 vs. the 20-bin baseline, and $\rho \geq 0.79$ against the Negative Log Calibration Score, a bin-free proper scoring rule.
- **Variance decomposition:** two-level and three-level ICC estimates match exactly — domain variance really is zero.
- **FDR threshold:** rejections hold at $\alpha = 0.01$ too: 55–57/57 for the four largest models, 46–55/57 for the smaller Qwens.

Takeaways

- **The unit of analysis is the subject, not the model.** A single aggregate scalar hides a 3.8–13.0× range in miscalibration magnitude — a single global fix (e.g. temperature scaling) can't work either: one scalar T tuned for the worst subjects (`professional_law`, `virology`, gap up to 0.487) would badly overcorrect well-calibrated subjects like `marketing` (gap 0.037).
- **Miscalibration is task-driven.** Subject rankings are stable across three families and six scales ($\rho \in [0.80, 0.94]$) — model-level retraining is unlikely to resolve it; the fix has to live at the subject level.
- **Practical upshot:** the recurring worst-calibrated subjects double as a portable pre-deployment checklist — flag likely miscalibration on a *new* model without re-running the full 57-subject audit.
- **Open question:** *why* these subjects are hardest (abstractness, training-data rarity, or something else) remains unresolved.
- **Caveats:** confidence is a single top-token logprob and may understate internal uncertainty post-RLHF; results are specific to MMLU's four-choice format; the Qwen scaling comparison spans only four checkpoints.
- **The audit generalizes:** the same multilevel-model + FDR pipeline applies to any benchmark with hierarchical structure — next steps include a fourth model family at matched scale and replication on MMLU-Pro.

References

Benjamini & Hochberg (1995). Controlling the false discovery rate. *JRSS-B* 57(1). Guo et al. (2017). On calibration of modern neural networks. *ICML*. Hendrycks et al. (2021). Measuring massive multitask language understanding. *ICLR*. Kadavath et al. (2022). Language models (mostly) know what they know. arXiv:2207.05221. Nakkiran et al. (2025). Trained on tokens, calibrated on concepts. arXiv:2511.04869. Xiong et al. (2024). Can LLMs express their uncertainty? *ICLR*.